

HW #1: Polynomial Calculator

Faculty of Computer and Information Science, Ben-Gurion University of the Negev

Spring 2026

Submission in pairs (groups of two students)

General Description

In this assignment you will implement a **Polynomial Calculator** using object-oriented design principles. The calculator supports arithmetic operations on polynomials whose coefficients are abstract scalar values.

The design must be general, extensible, and independent of any specific numeric type.

Concepts practiced

- Abstraction and interface design
- Inheritance and substitution
- Use of collections
- Clean and testable APIs

You are strongly encouraged to read the entire assignment before starting the implementation.

Scalar Abstraction

Scalar

Scalar represents an abstract numeric value. You may **not add data members** to this class.

API Specification

```
Scalar add(Scalar s)
Scalar mul(Scalar s)
Scalar neg()
Scalar power(int exponent)    // exponent >= 0
int sign()                   // 1, -1, or 0
boolean equals(Object o)     // override
String toString()            // override
```

IntegerScalar

Represents integer-valued scalars.

Data member

- private int number;

RationalScalar

Represents rational numbers.

Data members

- private int numerator;
- private int denominator;

API Specification

```
Scalar reduce()
```

Formatting rules

Let the rational value be $\frac{a}{b}$:

- If $\frac{a}{b}$ is an integer, return the integer
- Otherwise, return the reduced form a/b with no spaces
- Negative values must include a leading minus sign

Examples

```
12 / - 4 → "-3"  
-12 / - 5 → "12/5"  
12 / - 8 → "-3/2"
```

Note: Invalid rationals (e.g., denominator equal to zero) will not be tested.

Monomial

A **Monomial** represents a single polynomial term of the form

$$c \cdot x^k$$

where c is a Scalar and k is a non-negative integer.

Data members

- private int exponent;
- private Scalar coefficient;

API Specification

```
Monomial add(Monomial m)    // same exponent, otherwise null  
Monomial mul(Monomial m)  
Scalar evaluate(Scalar s)  
Monomial derivative()  
int sign()  
boolean equals(Object o)  
String toString()
```

Examples

Coefficient 1, exponent 3 $\rightarrow x^3$
Coefficient 4, exponent 1 $\rightarrow 4x$
Coefficient -1, exponent 7 $\rightarrow -x^7$
Any coefficient, exponent 0 $\rightarrow$ coefficient only

Polynomial

A **Polynomial** is defined as a collection of Monomials with distinct exponents.

Data member

- private Collection monomials;

API Specification

```
static Polynomial build(String input)

Polynomial add(Polynomial p)
Polynomial mul(Polynomial p)
Scalar evaluate(Scalar s)
Polynomial derivative()
boolean equals(Object o)
String toString()
```

The input string contains coefficients separated by spaces. The index of each coefficient corresponds to the exponent.

Examples

"1 2 3" $\rightarrow 1 + 2x + 3x^2$
"0 1 2 3" $\rightarrow x + 2x^2 + 3x^3$
"0 1/2 3 -5/3" $\rightarrow \frac{1}{2}x + 3x^2 - \frac{5}{3}x^3$

Additional Requirements

- Provide a UML class diagram
- Do not use casting (except where explicitly allowed)
- Do not use `instanceof` except inside `equals`
- Write unit tests for every class (at least one per method)
- Use packages; do not use the default package
- Helper methods are allowed, but make sure to keep the principles of object-oriented design in mind

Additional Questions

For additional questions, ask in the forum or during office hours. The responsible teaching assistant for this assignment is **Gilad Aharoni**.

Submission Instructions

Submit a single archive named:

`student1ID_student2ID.zip`

Archive contents

- **hw1.pdf** — UML class diagram
- **hw1.jar** — source code, tests, and compiled class files